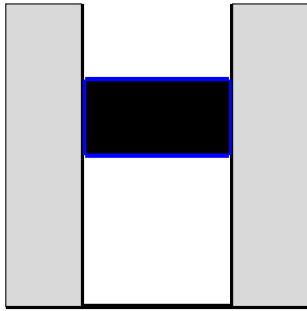


3. A block is held motionless between 2 walls

(a) Make a free body diagram for the block and the left and right walls. Assume none of the surfaces are frictionless



(b) Write Newton's Second Law for the block and the left and right walls

4. Below is a 2D cart with a vertical spring attached to a block. The cart also has 2 frictionless walls that hold the block motionless in the left and right directions relative to the cart. The system is currently at rest



(a) The cart is pushed to the right and the block is given an impulse in the downward direction both at $t = 0s$. Above, draw the theoretical trajectory of the block over time:

(b) What simple function could describe the motion of the block $x(t)$?